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Remote Recession Sensing of Ablative Heat Shield Materials

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Tests were performed to demonstrate the feasibility of a new method of measuring surface recession of a material sample during arc-jet testing in the NASA Ames mArc subscale developmental facility. The measurement principle was inspired through tracer elements such as Ca and Na which were seen in the spectra taken during the airborne observation campaign of the Stardust re-entry and which could be clearly observed standing out against the emission spectra emitted by post shock layer and glowing surface of the re-entry capsule. The measurement principle involves seeding of the heat shield materials at a defined depth with tracer elements which show strong and characteristic emission lines in the post shock plasma. Once the material recession reaches the seeding depth, these elements get into the hot plasma and show up in the emission spectra. The methodology was successfully demonstrated during arc-jet testing of phenolic impregnated carbon ablator (PICA) material which was seeded in depth with a mixture of NaCl and MgCl in powder form. In the emission spectroscopy data, the emission lines of Mg and Na showed up about 1.5s after probe insertion into the arc-jet plasma and vanished after another 2.5 seconds when recession had consumed the seeding material. From these data, a recession rate of about 1 mm/s is estimated. The heat flux during the test was measured to be 2575 W/cm^2 on a hemispherical heat flux probe which corresponds to a heat flux of 1036 W/cm^2 on the rectangular test articles. An estimate for a lower limit of the surface temperature of 2800K during the test was obtained by fitting Planck radiation to the continuum spectra emitted by the PICA sample. Typical recession rates of PICA during testing in the large arc-jet facilities at similar test conditions are reported to be on the order of 0.05 to 0.1cm/sec which agrees well with the recession rates of 0.05-0.06 cm/s estimated from the emission spectroscopy data. Further tests under better controlled conditions are suggested to quantify this measurement method. Through a different choice of seeding materials with lower melting point, an extension of the measurement principle to monitor char depth seems feasible but was not yet demonstrated. Possible applications besides ground testing are recession and possibly char depth measurements during real re-entry. Detection through emission spectroscopy could be accomplished through ground based or airborne observation as performed during the Stardust and Hayabusa re-entries, or through on-board spectrometers. A suitable mission would be the re-entry of the OSIRIS-REX mission planned for late 2023. The measured data are presented and interpreted, the results and details of future applications are discussed.

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I. Introduction

Material recession and charring are two major processes determining the performance of ablative heat shield materials. Even in ground testing, the characterization of these two mechanisms relies on measurements of material thickness before and after testing, thus providing only information integrated over the test time. For recession measurements, optical methods such as imaging the sample surface during testing are under investigation[1] but require high alignment and instrument effort, therefore being not established as a standard measurement method. For char depth measurements, the most common method so far consists in investigation of sectioned samples after testing or in the case of Stardust where core extractions were performed to determine char information.[2] In flight, no reliable recession measurements are available, except total recession after recovering the heat shield on ground. Developments of mechanical recession sensors have been started[5] but require substantial on board instrumentation adding mass and complexity. In this work, preliminary experiments to evaluate the feasibility of remote sensing of material recession and possibly char depth through optically observing the emission signatures of seeding materials in the post shock plasma is investigated. It is shown that this method can provide time resolved recession measurements without the necessity of accurate alignment procedures of the optical set-up and without any instrumentation on board of a spacecraft. Furthermore, recession data can be obtained without recovering flight hardware which would be a huge benefit for inexpensive heat shield material testing on board of small re-entry probes, e.g. on new micro-satellite re-entry probes as a possible future application of Cubesats[3] or RBR.[4]

In the spectrally resolved data gathered during the airborne observation campaign of the Stardust sample return capsule, the emission of tracer elements such as sodium, potassium, magnesium, and calcium was observed.[6] Calcium has been found to be present in the preform of the Phenolic Impregnated Carbon Ablator (PICA) forebody TPS material used on Stardust. Magnesium has been assigned to the protective paint of the heat shield. The strong appearance in the observed spectra was surprising, though, since these elements were not major constituents of the PICA material and are only present in trace concentrations. These results have been confirmed in preliminary ground testing of PICA in the NASA Ames arc-jet facilities.[7] Apparently, these tracer elements are able to diffuse against the incoming flow into the boundary layer and eventually into the post-shock layer where they emit substantial radiation due to the high plasma temperatures in this region. The concept of remote ablation sensing uses this effect by seeding the heat shield material at known depths with selected tracer materials. Once recession reaches the seeding depth, the characteristic emission of the seeding elements is expected to show up in the emission spectra emitted by the post-shock plasma. When the recession exceeds the maximum seeding depth, this emission signature should fade in the observed spectra. Since these signatures are distinguishes from the plasma emission in the spectral domain and due to the diffusion of the seeding material into the boundary and post-shock layer, no particular spatial resolution of the optical system is required as long as the majority of the post-shock system is covered by the collecting optics. A spatial resolution is obtained by restricting the seeding material to a selected position on the sample or heat shield. If the emission spectra are recorded continuously with time, a temporal assignment of a particular recession depth (pre-selected through seeding a defined in-depth position) is possible. To demonstrate this concept, tests with different seeding materials and configurations were done in the Miniature Arc Jet (mARC) facility at NASA Ames.[8]

II. The mARC Facility

The mARC is a subscale segmented arc heater in which a continuous electrical discharge between electrodes in the arc chamber heats and expands air at a mass flow rate of approximately 0.45 g/s (in this study) to high temperatures and supersonic velocities. The DC arc jet is composed of a cylindrically symmetric geometry consisting of a coolant-cooled thoriated tungsten tipped cathode, a water-cooled copper anode, a constrictor channel, and a converging-diverging nozzle. A schematic of the cross section of the mARC can be seen in Figure 1. In operation, a Hypertherm Max 200 30 kW plasma cutting power supply capable of supplying up to 200 A DC establishes a high current regulated arc. The arc is directed from the cathode tip, through a 1.3 cm diameter constricted arc column channel, and attaches to the anode. Air is swirled into the arc column through injection ports located behind the cathode. A swirling ring is used to stabilize the arc, constrain the hot gas discharge column to the axis of the vortex, and bring the gas into longer and more effective contact with the arc. The resultant constricted arc heats and largely ionizes the air on the centerline of the constrictor and continues to heat the boundary air through radiation and conduction, resulting in a large radial temperature gradient. It is this large radial temperature gradient which allows the centerline temperature of the heated air to be very high without melting the nozzle. After being directed through the arc column, the hot air is then accelerated out through the convergent-divergent nozzle (1 cm exit diameter) forming a plasma column approximately 1.3 cm in diameter.

The mARC uses a cylindrical vacuum chamber of approximately 0.61 m in length and 0.3 m diameter connected to two Leybold Trivac (model # D30A) rotary vane dual stage mechanical vacuum pumps capable of providing base pressures in the 0.001 kPa (10 mTorr) range. Chamber background pressures consistently reached approximately 2.67 kPa (20 Torr) during operation. The mARC anode was water cooled by a Goulds Pumps e-SV Series vertical multi-stage pump capable of flowing water at 7.885 kg/s. A cooled diffuser was placed at the lower end of the chamber to prevent overheating of the chamber walls due to the high flow temperatures. The layout of the facility can be seen in Figure 2.

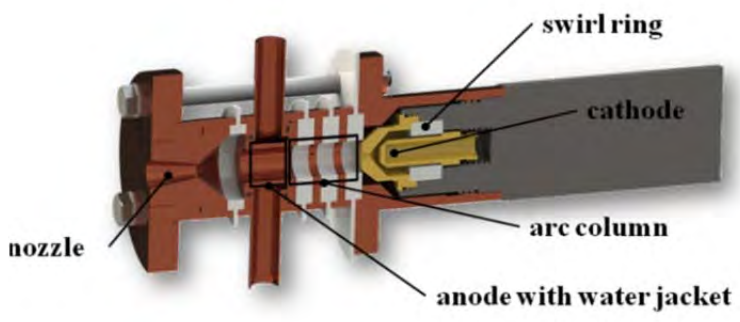

Figure 1: Cross sectional schematic of the mARC[8]

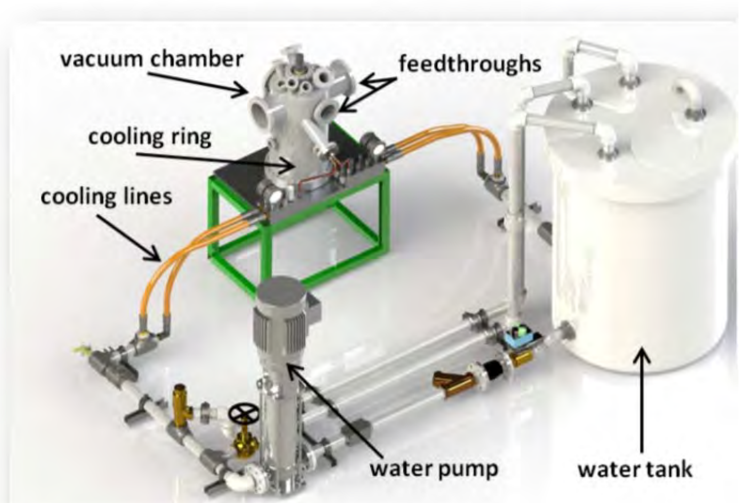

Figure 2: The mARC vacuum facility[8]

III. Calibration Runs

For this particular test campaign, the plasma cutting power supply operated at 70 A DC. Prior to inserting PICA test coupons at the center of the plasma stream at approximately 5mm from the nozzle exit, calibration runs were performed to obtain stagnation heat flux at the same axial location by using a 4.76 mm (0.187 inch) diameter hemispherical water-cooled Gardon Gage (Serial Number: 180241) from Medtherm Corporation. Refer to Figure 3 for additional dimensions of the calorimeter.

Due to extreme heating environment and thermal limitation of the material, the exposure time of the Gardon Gage in the flow was short. A sample profile produced by the calorimeter is shown in Figure 4, and an estimated heat flux was obtained by averaging the data on a small time interval.

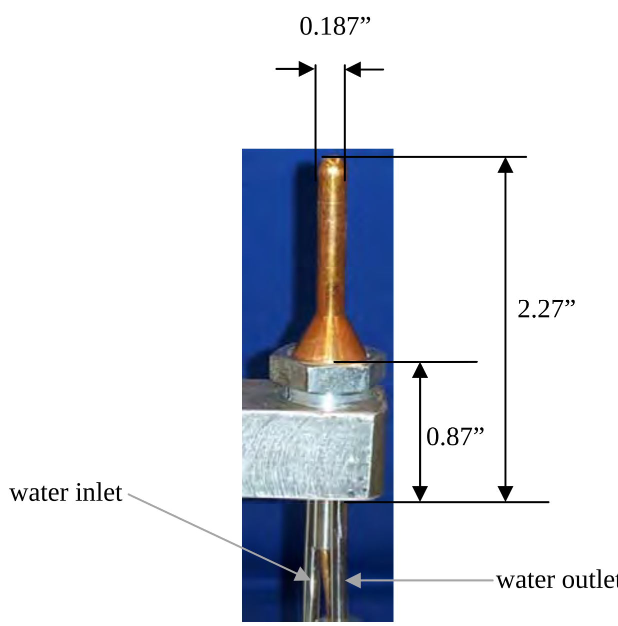

Figure 3: Dimensions and mounting of Gardon Gage.

This technique was applied to all calibration runs with similar heat flux profiles and the values are shown in Table 1, which indicates the thermal condition which the test coupons were exposed to and that the measurements were repeatable.

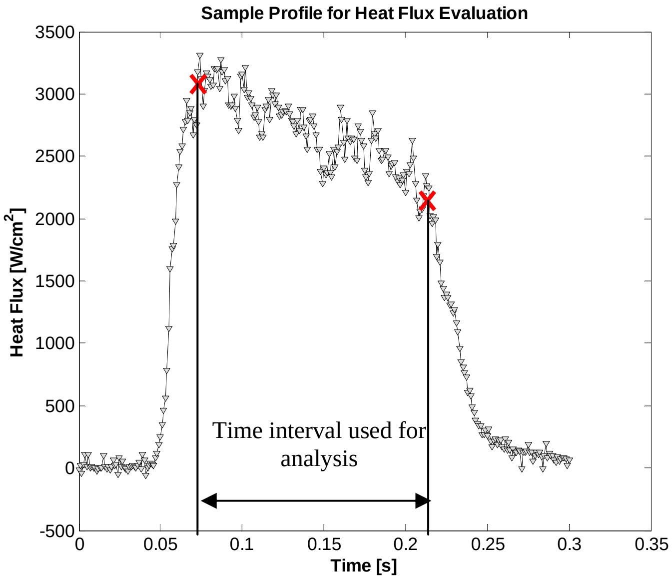

Figure 4: Sample calorimeter output for heat flux evaluation.

Table 1: Summary of stagnation heat flux for calibration runs.

Test #	Heat Flux [W/cm^2]
1	2500
2	2600
3	2600
4	2600

Due to the uncertainties of the heat flux measurements of ±10%, the average of these values of 2575 W/ cm^2 will be used for the following considerations. The heat flux to the flat face, rectangular sample is determined using the relations according to *Zobey and Sullivan*.[11] For a flat face sample, the ratio of flat face radius r_B to radius of curvature r_N is zero. If the ratio of corner radius r_C to flat o 0.285 and the effective radius becomes r_{eff}=14.7 mm. Therefore, the flat face heat flux becomes

$$\dot{q}_{BB} = \dot{q}_{Hemi}\sqrt{\frac{r_{Hemi}}{r_{BB}}} = 2575\sqrt{\frac{2.38}{14.7}}\frac{W}{cm^2} = 1036\frac{W}{cm^2} \quad .$$

IV. Sample Geometry and Seeding Procedure

PICA was chosen as the baseline material to demonstrate the seeding concept as it is a heritage TPS material and the baseline forebody heatshield for OSIRIS REx. Additionally, PICA is a low-density material that is easily machined making it suitable for this demonstration experiment, and PICA has limited constituents thereby not adding further to the emission spectra. Two sets of PICA coupons were evaluated; 6.35mm (0.25") diameter cylindrical (Figure 5) and 8.38mm (0.33") rectangular coupons (Figure 6). The smaller coupons were more challenging to align during testing, however the larger coupons were successfully tested.

Figure 5: (a) PICA cylindrical test coupons (6.35mm diameter) with seeded materials inserted in the PICA prior to encapsulating

A series of seeding materials were evaluated including, metals, oxides and salts of Al, Ti, Mg and Si. Sodium chloride was also included. All seeding materials were in powder form having a high surface area (mean powder diameter of a few microns).

The seeding materials were extruded with a polymer matrix to form seeding rods. The PICA coupons were drilled at given locations and the seeding rods were inserted into the PICA and sealed using a PICA like paste allowing the seeding material to be encapsulated within the PICA. An example of the seeded materials inserted in the PICA prior to encapsulating is shown in Fig. 4 for the cylindrical samples. Pre- and post-test pictures of the rectangular coupons are shown in Fig. 5. Some samples showed asymmetric recession patterns which is assigned to a slightly off-centered position of the sample in the plasma. Due to limited test time not all seeding materials were evaluated in the mARC facility. The results presented will focus on samples seeded with NaCl and MgCl.

(a) **(b)** **(c)**

Figure 6: Typical PICA rectangular test coupons (a) pre test, (b) during test, and (c) post test.

All samples were kept in the flow for 10 seconds. From the total recession of the sample and the test time, an average total recession rate was calculated. For the sample shown in Fig. 6, the total recession rate obviously varies over the surface, at the center of the sample, however, the recession seems mostly constant. For future comparisons the highest recession values of about 0.056 cm/s (determined as the average value from the right side of the sample in Fig. 6 to about ¾ of the sample width) are used for a comparison with values determined from the optical measurements and with typical recession data in the IHF arc-jet facility.[10]

V. Optical Set-up and Results

The emission was taken using a focal point imaging system with a 50 mm (2 inch) spherical mirror with a focal length of 305 mm (12 inch) providing a mostly collimated beam with the mirror diameter. The light was focused on a 200μm optical fiber and detected with an miniaturized spectrometer (Avantes AvaSpec-3648-USB2-UA, nominal wavelength range 180-1100nm). The f number of the fiber (numerical aperture n_A=0.22) was matched using an aperture between mirror and fiber. The beam cross section covered the arc-jet nozzle exit with its upper edge, the virgin sample surface placed approximately at 10mm from the upper edge of the detection beam. The alignment was done using a HeNe laser which was reversely fed into the collimating system and therefore illuminated the field of view of the spectrometer as shown in Fig. 7.

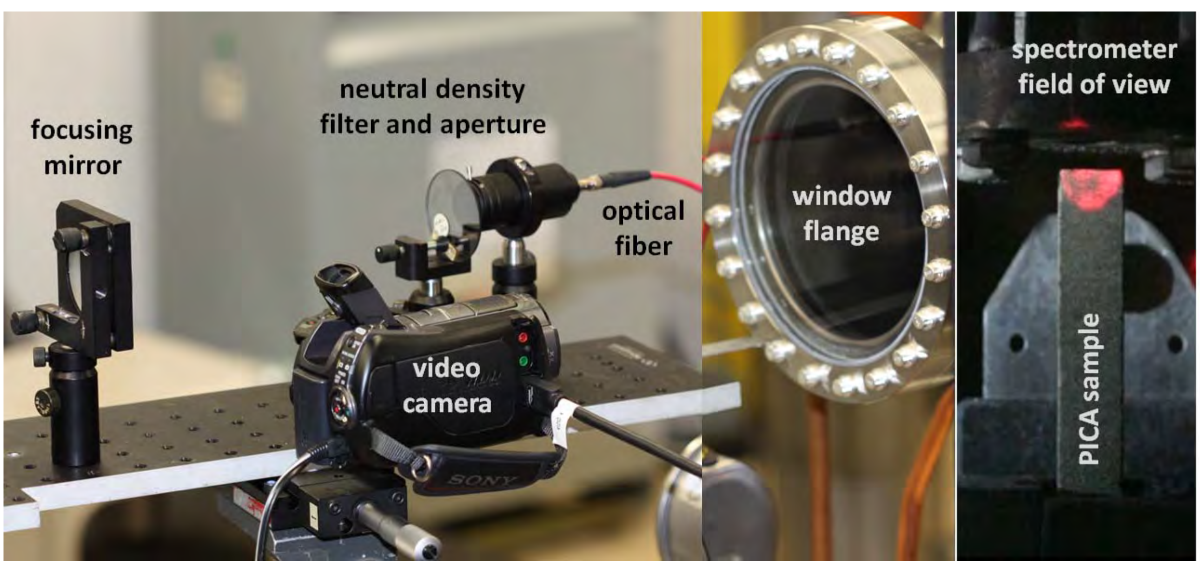

Figure 7: Optical set-up with spectrometer field of view illuminated by a HeNe laser.

During recession, the surface did move through the beam center towards the lower edge. In pretests, the set-up sensitivity over the beam cross section was found to change moderately depending on the fiber orientation. Therefore, it cannot to be excluded that the sensitivity to light coming from the receding surface might change with test time, though considered minor due to the experience during the pretests. The measurements were calibrated to absolute intensities using an integrating sphere.

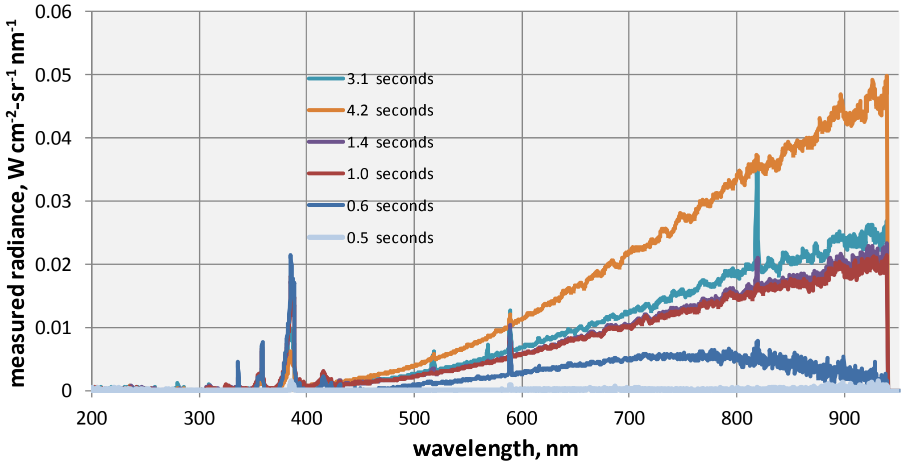

Figure 8: Selected emission spectra for increasing test time.

Data acquisition was started shortly before the sample was inserted with acquisition times of 15ms and a data rate of ~67Hz. During analysis, data were averaged to one point every 0.105 seconds. Background noise was measured separately after the test and was subtracted from the measured data. All data were wavelength calibrated through measurements of a Hg calibration lamp, and intensity calibrated to spectral radiance (W cm^{-2} sr^{-1} nm^{-1}) through measurements of a halogen integrating sphere and a Deuterium lamp. Synthetic spectra of Mg and Na were generated based on NIST tabulated spectroscopic data using Boltzmann distributions and a series of temperatures. The data shown for comparison were generated using Boltzmann distributions of the electronically excited levels for a temperature of 9000K which was arbitrarily chosen to show also weaker lines in the spectrum and is not considered a valid plasma temperature.

The continuum emission of the glowing probe showed an increase with test time as shown in Fig. 8 which is assigned to a change in visible glowing surface area of the wide field detection optics. CN emission around 390 nm was detected throughout the whole test with decreasing intensity with test time. Emission lines of Mg (280nm, 285nm, and 518nm) and Na (568nm, 589nm, and 819nm) were identified in the arc-jet plasma. The time trace of the spectral emission is shown in Fig. 8, the identification of the Mg lines is demonstrated in Fig. 9.

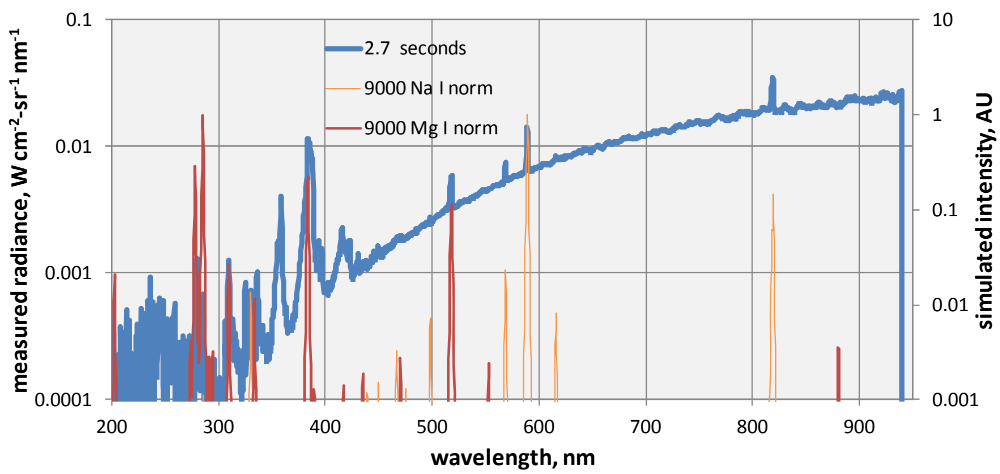

Figure 9: Measured spectra at about 2.3 seconds after sample insertion in comparison to synthetic spectra of sodium and magnesium derived from NIST tabulated spectroscopic constants.

All identified lines started showing up about 1 second after sample insertion and disappeared at roughly 4 seconds after insertion, as illustrated in Fig. 8 by the integrated line emissions. Only the Na line at 589 nm was present right after sample insertion, however, it disappeared together with the other emission lines. The fact that only this individual Na line showed up at the beginning of the test and not the ones observed later indicates a significantly lower excitation temperature for the sodium emission at the beginning of the test in comparison to the later emission. The latter one seems clearly related to ablation processes (as indicated by the other observed atom lines). It might be that residual sodium from the ambient air is responsible for the early Na emission. Another explanation might be sodium on the sample surface which would explain why the emission shows a local maximum right at the beginning followed by a decreasing trend.

As shown in Fig. 6, the surface recession over the sample was not entirely uniform, showing stronger total recession at the center and lower values at the edges. For a first estimate of recession rate from the emission spectroscopy data, the time within which the Mg emission at 518 nm was observed with more than 20% of the maximum strength of Δt=2.4 s was chosen to indicate the time the seeding material was exposed to the plasma (compare Fig. 10). If the diameter of the hole filled with seeding material of 1.5 mm is chosen as thickness of the seeding layer and interpreted as the surface recession during which Mg emission was seen, a recession rate of 0.66 mm/s is obtained. If the first and last appearance of Mg emission of Δt=3.5 s is used instead, a minimum recession rate of 0.44 mm/s is obtained from the emission spectroscopic measurements. From post test photographs, the recession rate in the center region of the sample was estimated between 0.5 and 0.6 mm/s. Given the asymmetric recession of the probe this agreement is considered rather good.

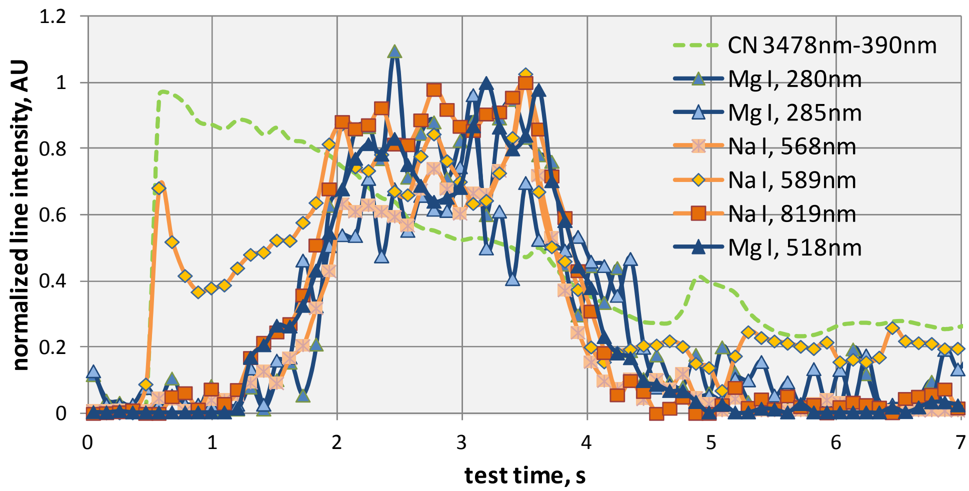

Figure 10: Time trace of spectrally integrated line emission of sodium and magnesium.

In Fig. 11, these recession rates are plotted vs. sample temperature and are compared with data published by Covington et al. obtained during testing in the 60 MW IHF arc-jet facility . In these measurements, the operating condition of the arc-jet was held constant at a bulk enthalpy level of 29.5 MJ/kg and flat face models of different diameter were used, yielding different heat flux levels. The recession rates determined in the mArc test campaign are in rather good agreement with the IHF data at comparable heating rates, although the sample temperature in the IHF was higher by about 10%. This, however, may be explained with the bad visibility of the sample surface for the optical path pf the spectrometer which only allowed for a measurement of an averaged sample temperature, dominated by the sample side and not by the surface. Taking this into account, reasonable agreement with typical PICA recession during arc-jet testing is obtained.

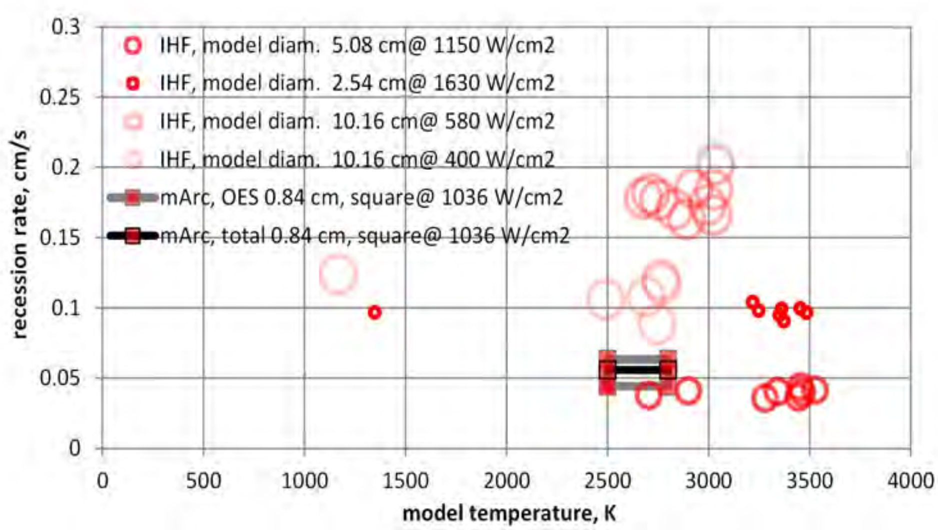

Figure 11: Recession rates of PICA during mArc testing compared to data obtained in the NASA Ames IHF arc-jet facility[10] vs. sample temperature.

The continuum part of the emission spectra, emitted by the glowing PICA material was used to estimate the temperature of the sample during the test. As already shown in Fig. 7, the continuum emission showed a continuous increase with test time. Since the field of view of the spectrometer was chosen larger than the sample surface to cover the whole post-shock region, this increase is explained with a larger glowing area visible in that field of view as the sample recessed. The quantity of measured intensity is therefore not suitable for a temperature determination. Thus, the emission spectra were normalized to their intensity in a wavelength region between 600 nm and 700 nm which did not show emission of the seeding elements or of plasma emission. After normalization, all spectra basically collapse to one continuum spectrum for the first 4 seconds of test time which indicates that the average sample temperature was fairly constant during that period. This spectrum was compared to Planck radiation which was normalized in the same wavelength region. An emissivity value of 0.9 was used as reported for charred PICA[9,10]and used for the interpretation of flight data.[6] The best agreement between Planck radiation and experimental spectra was found for a temperature of 2500K during the first four seconds of test time. During the remaining six seconds of test time, the spectral shape transformed to higher Planck temperatures of about 2800 K at 10 seconds close to the end of the test. Since the spectrometer was set up perpendicular to the arc-jet flow, the front surface of the sample was barely visible in the spectrometer field of view. As the sample recessed, more and more of the front surface became visible which explains the increasing Planck temperature towards the end of the test. However, even at the end of the test, the spectrometer had still not a clear view of the sample front surface. Therefore, these temperatures are interpreted as lower limits of the front surface temperature which may be significantly higher than the values determined from the emission spectra. The normalized Planck radiation for temperatures between 2400K and 2900K are shown in Fig. 12 in comparison to the experimental spectra. The variation in spectral shape of the computed Planck radiation suggests an accuracy of this temperature of at best ±50 K.

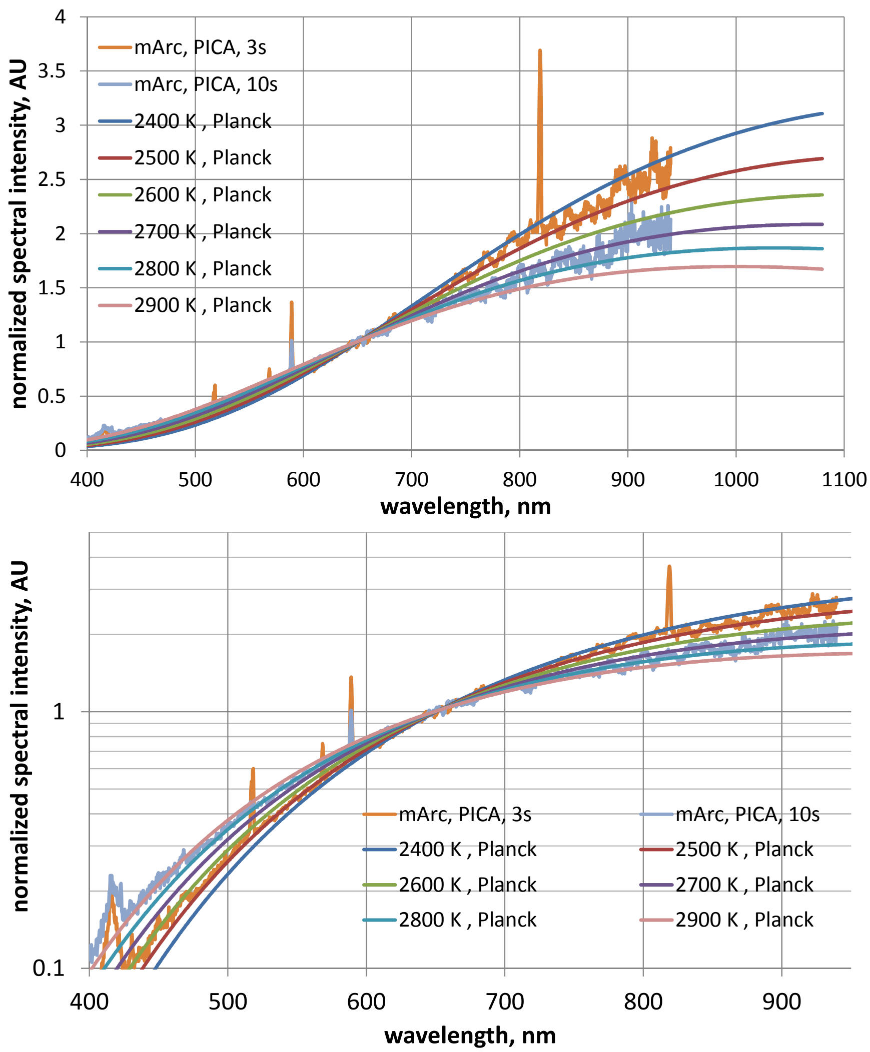

Figure 12: Measured spectra at 3 and 10 seconds after sample insertion in comparison to Planck radiation at temperatures between 2400 K and 2900 K on linear and logarithmic scale. All spectra are normalized to the emission between 600 nm and 700 nm.

VI. Conclusions

The tests in the mARC facility demonstrated the feasibility of a remote ablation sensor during arc-jet testing through in-depth seeding and emission spectroscopy during testing. From heat flux measurements with a hemispherical probe the corresponding heat flux on the flat face sample surface was determined to 1036 W/cm^2. First emission signatures of the seeding elements (Mg and Na) were detected about 1.2 seconds after the article was inserted into the plasma flow and faded out after about 3 seconds. The 589 nm emission line of Na was present during the whole test time, though, which may be assigned to residual sodium in the ambient air. These values yield a recession rate of about 0.05-0.06 cm/s which agrees well with the total recession rate of the sample determined after the test, and seems reasonable if compared with recession rates measured in the IHF arc-jet facility at corresponding heat flux levels. From the spectral shape of the continuum emission, the sample front surface is anticipated to have been at temperatures in excess of 2800K which also seems reasonable if compared to typical conditions during IHF testing at comparable heat loads. For future measurement campaigns, parametric studies will have to be conducted to determine optimum seeding amounts and strategies as well as further suitable seeding materials. For these investigations, small plasma facilities are well suited to avoid high testing costs. Once suitable seeding materials and their emission signatures are sufficiently characterized, testing in qualification facilities capable to produce re-entry relevant conditions and spatially and temporarily constant recession rates are recommended to quantitatively investigate the sensing method and relate the results to traditional recession measurements. It seems feasible to extend this measurement to not only measure recession but also char depth by using combinations of seeding elements with different melting points. The material with a low melting point (e.g. aluminum) is anticipated to evaporate with the pyrolysis gases once the pyrolysis zone reaches the seeding depth and could be detected once in contact with the plasma. As soon as the recession reaches the seeding depth, the material with high melting points will show up in the emission spectra as demonstrated in the presented work.

It should be mentioned that the optical set-up used in this work did not require any spatial resolution of the seeding area but monitored the whole post-shock region. Spatial resolution of the optical set-up is not required. One attractive application of this method is the seeding of re-entry capsule heat shields to gather in-flight data using airborne or ground based observation of the re-entry, one desirable candidate mission being the upcoming OSIRIS-Rex mission by NASA Goddard and Lockheed Martin. However, small re-entry probes for inexpensive flight testing of heat shield materials would be a suitable application, too, since remote sensing would enable data acquisition of heat shield performance and recession without the need of recovering the flight hardware. If no external observation is possible, the emission of tracer elements might as well be performed through flight hardware since the emission of the tracer elements is generated in the post-shock region which could be monitored from the inside of the entry vehicle.

Acknowledgments

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